

SOFTWARE REQUIREMENTS DOCUMENT

Bakery Reseller Management Application

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Section 1 System Overview

1.1 Purpose

This document specifies the functional and non-functional requirements for the Bakery Reseller Management Application (BRMA). It is intended for use by software engineers, QA teams, and product managers responsible for designing, building, testing, and maintaining the system.

1.2 System Description

BRMA is a mobile-first operations and financial management platform that enables a bakery company to track bread distribution to resellers, record daily financial settlements, detect anomalies, and generate operational and financial reports. The system covers the full lifecycle of each reseller's daily cycle: morning stock distribution, evening cash reconciliation, and monthly balance settlement.

1.3 Scope

The system covers the following operational domains:

- Morning bread distribution management
- Evening cash-in and signed-amount recording
- Per-reseller daily financial balance calculation
- Cumulative monthly balance tracking and payout computation
- Anomaly and inconsistency detection
- Role-based reporting and dashboards

Out of Scope (v1.0)

Inventory procurement, supplier management, payroll processing, and customer-facing sales interfaces are explicitly excluded from this version.

1.4 Intended Users

- Depot Manager : primary daily operational user, mobile-device operator
- Owner / Admin : oversight, reporting, and configuration authority

Section 2 Actors and Roles

The system defines two user roles. Access permissions are strictly role-bound and non-interchangeable without explicit administrative grant.

Role	Primary Interface	Responsibilities
Depot Manager	Mobile (Android / iOS)	Record distributions, record payments, record signed amounts, review daily summaries
Owner / Admin	Desktop + Mobile	View all reports, configure resellers and products, approve monthly payouts, set bonuses, review anomalies

2.1 Role Permissions Matrix

Permission / Action	Depot Manager	Owner / Admin
Record bread distribution	✓	✓
Record cash payment (Amount Paid)	✓	✓
Record signed amount (Amount Signed)	✓	✓
View own daily summary	✓	✓
View all resellers' summaries	X	✓
View financial reports	X	✓
Configure resellers	X	✓
Configure products and prices	X	✓
Set bonus amounts	X	✓
Approve monthly payouts	X	✓
Acknowledge anomaly alerts	✓	✓
Export data / reports	X	✓

Section 3 Business Workflow Description

3.1 Morning Process — Bread Distribution

Executed each operational morning by the Depot Manager:

1. The Depot Manager receives the day's bread stock from the bakery production unit.
2. For each reseller, the Depot Manager opens a new distribution record for the current date.
3. For each product distributed to the reseller, the Depot Manager enters:
 - Reseller identifier
 - Product name / SKU
 - Quantity distributed (integer units)

- Unit company price (wholesale price, in local currency)
- 4. The system automatically calculates and stores:
 - Expected Company Amount = Quantity × Unit Company Price
- 5. The distribution record is saved and locked from further quantity edits after the manager confirms dispatch (subject to admin override only).

Critical Business Rule — No Returns

Once bread is distributed to a reseller, it is the reseller's sole responsibility. The company does NOT accept return of unsold bread. The reseller remains liable for the full wholesale value of all distributed units, regardless of actual sales performance.

3.2 Evening Process — Cash Settlement

Executed each operational evening by the Depot Manager upon receiving cash from each reseller:

6. The reseller returns cash to the Depot Manager.
7. The Depot Manager opens the reseller's record for the current day.
8. The Depot Manager enters the Amount Paid (the total cash physically returned by the reseller.)
9. If the reseller requests some money back (for personal expenses or other purposes), the Depot Manager records the Amount Signed - the cash given back to the reseller from the collected funds.
10. The system calculates the following settlement values:
 - Cash After Signing = Amount Paid - Amount Signed
 - Account Balance = Cash After Signing - Company Due
11. The system displays the Account Balance interpretation to the Depot Manager:
 - Positive balance: reseller has savings held by the company
 - Zero balance: reseller is fully settled for the day
 - Negative balance: reseller owes money to the company
12. The evening record is saved. If Amount Signed is not entered, it defaults to 0.00.

3.3 Monthly Accumulation and Payout Process

13. Throughout the month, the system accumulates each reseller's daily Account Balance values.
14. The running cumulative balance is visible to the Admin at any time.
15. At month-end, the Admin reviews each reseller's Net Monthly Balance.
16. The Admin may optionally assign a Bonus amount per reseller.
17. The system calculates: Final Payout = Net Monthly Balance + Bonus
18. The Admin reviews and confirms each payout.

19. Upon confirmation, the payout record is marked as settled and the reseller's cumulative balance resets to 0.00 for the new month.

Section 4 Functional Requirements

4.1 Module: User Authentication and Access Control

- **FR-AUTH-01** | The system shall require username and password authentication for all users before granting access to any feature.
- **FR-AUTH-02** | The system shall enforce role-based access control strictly as defined in Section 2.1.
- **FR-AUTH-03** | User sessions shall expire after 8 hours of inactivity.
- **FR-AUTH-04** | The Admin shall be able to create, deactivate, and reset credentials for Depot Manager accounts.
- **FR-AUTH-05** | All authentication events (login, logout, failed attempts) shall be logged with timestamp and device identifier.

4.2 Module: Reseller Management

- **FR-RES-01** | The Admin shall be able to create a reseller profile containing: full name, phone number, assigned zone/route, and status (active/inactive).
- **FR-RES-02** | The system shall prevent deletion of reseller records that have associated financial history; deactivation shall be used instead.
- **FR-RES-03** | The Admin shall be able to view the complete financial history of any reseller at any time.
- **FR-RES-04** | Reseller records shall be searchable and filterable by name, zone, and status.

4.3 Module: Product and Price Management

- **FR-PROD-01** | The Admin shall be able to create and manage products with the following attributes: product name, SKU, default company unit price, and status (active/discontinued).
- **FR-PROD-02** | The system shall maintain a price history log for each product; price changes shall not retroactively alter past distribution records.
- **FR-PROD-03** | The Depot Manager shall be able to override the unit company price at the time of distribution entry, subject to a confirmation prompt.
- **FR-PROD-04** | Price values shall support up to two decimal places in local currency.

4.4 Module: Distribution Management (Morning)

- **FR-DIST-01** | The Depot Manager shall be able to create a distribution record per reseller per day.

- FR-DIST-02 | A single distribution record may contain multiple product lines (one row per product).
- FR-DIST-03 | For each product line, the system shall record: reseller ID, date, product ID, quantity, and unit company price.
- FR-DIST-04 | The system shall automatically calculate and persist: Expected Company Amount = Quantity × Unit Company Price.
- FR-DIST-05 | The system shall prevent duplicate distribution records for the same reseller on the same date unless explicitly overridden by the Admin.
- FR-DIST-06 | Once the Depot Manager confirms dispatch, quantity fields shall be locked. The Admin may unlock a record for correction, with mandatory audit log entry.
- FR-DIST-07 | The Depot Manager shall be able to view a summary of total bread distributed by product and total expected company amount for the current day.

4.5 Module: Payment Settlement (Evening)

- FR-PAY-01 | The Depot Manager shall be able to open any reseller's daily record and enter Amount Paid and Amount Signed.
- FR-PAY-02 | Amount Signed shall default to 0.00 if not entered.
- FR-PAY-03 | The system shall automatically calculate and persist: Cash After Signing = Amount Paid – Amount Signed.
- FR-PAY-04 | The system shall automatically calculate and persist Account Balance as defined in Section 7.
- FR-PAY-05 | The system shall display the balance interpretation (positive / settled / negative) immediately upon save.
- FR-PAY-06 | Amounts shall not accept negative input values; the system shall validate and reject such entries with an error message.
- FR-PAY-07 | Once the evening record is submitted, it shall be locked. Admin override required for corrections, with mandatory audit log entry.

4.6 Module: Monthly Balance and Payout

- FR-MONTH-01 | The system shall maintain a running cumulative balance for each reseller, updated daily.
- FR-MONTH-02 | The Admin shall be able to view each reseller's cumulative balance at any point during the month.
- FR-MONTH-03 | The Admin shall be able to assign an optional bonus amount per reseller before finalising the monthly payout.
- FR-MONTH-04 | The system shall calculate: Final Payout = Net Monthly Balance + Bonus.
- FR-MONTH-05 | The Admin shall confirm each payout before it is recorded as settled.
- FR-MONTH-06 | Upon payout confirmation, the system shall: record the payout transaction with date and amount, reset the reseller's cumulative balance to 0.00, and mark the monthly period as closed.
- FR-MONTH-07 | Closed monthly periods shall be read-only. Corrections require a documented administrative adjustment entry.

4.7 Module: Audit Trail

- FR-AUDIT-01 | Every create, update, delete, and override action on financial records shall be logged with: user ID, timestamp, action type, affected record ID, and before/after field values.
- FR-AUDIT-02 | Audit logs shall be read-only to all users; they shall not be editable or deletable.
- FR-AUDIT-03 | The Admin shall be able to filter and export audit logs by date range, user, and action type.

Section 5 Non-Functional Requirements

5.1 Performance

- NFR-PERF-01 | All screen loads and data submissions shall respond within 2 seconds under normal network conditions (3G or above).
- NFR-PERF-02 | Financial calculations (balance, payout) shall complete server-side within 500 milliseconds.
- NFR-PERF-03 | The system shall support at least 50 concurrent users without degradation.

5.2 Reliability and Availability

- NFR-REL-01 | The system shall have a minimum uptime of 99.5% during operational hours (05:00–22:00 local time).
- NFR-REL-02 | Data entered by the Depot Manager shall be persisted within 3 seconds of submission to prevent loss on connectivity interruption.
- NFR-REL-03 | The system shall support offline data entry for distributions and payments, synchronising automatically when connectivity is restored.

5.3 Security

- NFR-SEC-01 | All data in transit shall be encrypted using TLS 1.2 or higher.
- NFR-SEC-02 | Passwords shall be stored using a cryptographic hash function with salt (minimum bcrypt cost factor 12).
- NFR-SEC-03 | The system shall implement rate-limiting on login attempts: maximum 5 failed attempts before a 15-minute lockout.
- NFR-SEC-04 | Financial records shall not be accessible via unauthenticated API endpoints under any circumstances.

5.4 Usability

- NFR-USE-01 | The mobile interface shall be operable with one hand on standard smartphone screen sizes (360×640 dp minimum).

- NFR-USE-02 | The Depot Manager's most frequent actions (new distribution, record payment) shall be reachable within 2 taps from the home screen.
- NFR-USE-03 | All numeric input fields shall display a numeric keyboard by default on mobile devices.
- NFR-USE-04 | Error messages shall be specific and actionable (e.g., 'Amount Paid cannot be empty' rather than 'Invalid input').

5.5 Scalability and Maintainability

- NFR-SCAL-01 | The data model shall accommodate growth to 500 active resellers and 5 years of historical records without architectural changes.
- NFR-SCAL-02 | Product catalogue and price structures shall be configurable without code changes.
- NFR-SCAL-03 | The system shall be built using a modular architecture to facilitate the future addition of features such as route management, SMS notifications, and accounting integrations.

5.6 Data Integrity

- NFR-INT-01 | All financial calculations shall be performed server-side and stored persistently; client-side calculations shall be used for display preview only.
- NFR-INT-02 | Currency values shall be stored as integers (smallest currency unit, e.g., centimes) to prevent floating-point rounding errors.
- NFR-INT-03 | Database transactions shall be used for all multi-record financial writes to ensure atomicity.

Section 6 Data Model

The following entities constitute the core data model. Fields marked * are mandatory.

6.1 Entity: User

Field	Type	Constraints	Description
user_id *	UUID	PK, unique	System-generated identifier
username *	VARCHAR(50)	Unique, not null	Login username
password_hash *	VARCHAR(255)	Not null	bcrypt hashed password
full_name *	VARCHAR(100)	Not null	Display name
role *	ENUM	depot_manager admin	User role
is_active *	BOOLEAN	Default: true	Account status

Field	Type	Constraints	Description
created_at *	TIMESTAMP	Auto-set	Account creation time
last_login	TIMESTAMP	Nullable	Last successful login

6.2 Entity: Reseller

Field	Type	Constraints	Description
reseller_id *	UUID	PK, unique	System-generated identifier
full_name *	VARCHAR(100)	Not null	Reseller's legal name
phone	VARCHAR(20)	Nullable	Contact phone number
zone	VARCHAR(50)	Nullable	Assigned market zone/route
is_active *	BOOLEAN	Default: true	Operational status
created_at *	TIMESTAMP	Auto-set	Record creation time
notes	TEXT	Nullable	Admin notes

6.3 Entity: Product

Field	Type	Constraints	Description
product_id *	UUID	PK, unique	System-generated identifier
name *	VARCHAR(100)	Not null	Product display name
sku	VARCHAR(30)	Unique, nullable	Stock-keeping unit code
default_unit_price *	INTEGER	Not null, > 0	Default company price (smallest currency unit)
is_active *	BOOLEAN	Default: true	Active / discontinued status
created_at *	TIMESTAMP	Auto-set	Record creation time

6.4 Entity: Distribution Record

Field	Type	Constraints	Description
distribution_id *	UUID	PK, unique	System-generated identifier
reseller_id *	UUID	FK → Reseller	Linked reseller
date *	DATE	Not null	Distribution date
product_id *	UUID	FK → Product	Distributed product
quantity *	INTEGER	Not null, > 0	Units distributed
unit_company_price *	INTEGER	Not null, > 0	Price at time of distribution

Field	Type	Constraints	Description
expected_company_amount *	INTEGER	Calculated, stored	quantity × unit_company_price
is_locked *	BOOLEAN	Default: false	Locked after dispatch confirmation
created_by *	UUID	FK → User	Depot manager who created record
created_at *	TIMESTAMP	Auto-set	Record creation time

6.5 Entity: Daily Settlement Record

Field	Type	Constraints	Description
settlement_id *	UUID	PK, unique	System-generated identifier
reseller_id *	UUID	FK → Reseller	Linked reseller
date *	DATE	Not null	Settlement date
company_due *	INTEGER	Calculated, stored	Sum of expected amounts for the day
amount_paid *	INTEGER	Not null, ≥ 0	Cash returned by reseller
amount_signed *	INTEGER	Not null, ≥ 0	Cash taken back by reseller
cash_after_signing *	INTEGER	Calculated, stored	amount_paid – amount_signed
account_balance *	INTEGER	Calculated, stored	cash_after_signing – company_due
is_locked *	BOOLEAN	Default: false	Locked after submission
created_by *	UUID	FK → User	Depot manager who created record
created_at *	TIMESTAMP	Auto-set	Record creation time

6.6 Entity: Monthly Payout Record

Field	Type	Constraints	Description
payout_id *	UUID	PK, unique	System-generated identifier
reseller_id *	UUID	FK → Reseller	Linked reseller
period_year *	SMALLINT	Not null	Fiscal year
period_month *	SMALLINT	1–12, not null	Fiscal month
net_monthly_balance *	INTEGER	Calculated, stored	Sum of all daily account_balance values
bonus	INTEGER	Default: 0, ≥ 0	Optional bonus set by Admin

Field	Type	Constraints	Description
final_payout *	INTEGER	Calculated, stored	net_monthly_balance + bonus
status *	ENUM	pending confirmed	Payout workflow state
confirmed_by	UUID	FK → User, nullable	Admin who confirmed payout
confirmed_at	TIMESTAMP	Nullable	Confirmation timestamp

Section 7 Financial Rules and Formulas

All financial formulas are authoritative. All calculations shall be performed server-side and results persisted to the database. Client-side values are for display preview only and shall never be used as stored values.

7.1 Distribution Calculation

Expected Company Amount = Quantity × Unit Company Price

Example: 50 loaves × 250 XAF = 12,500 XAF

7.2 Daily Settlement Calculations

Company Due = SUM(Expected Company Amount) for all products on date D for reseller R

Cash After Signing = Amount Paid - Amount Signed

Account Balance = Cash After Signing - Company Due

7.3 Account Balance Interpretation

Condition	Meaning	System Action
Account Balance > 0	Reseller has overpaid. The surplus is savings held by the company.	Display as positive savings. Accumulate into monthly balance.
Account Balance = 0	Reseller is fully settled for the day.	Display settled status. No further action required.

Condition	Meaning	System Action
Account Balance < 0	Reseller owes money to the company for the day.	Display outstanding debt. Flag if deficit exceeds threshold (configurable). Accumulate into monthly balance.

7.4 Monthly Accumulation and Payout

Net Monthly Balance = SUM(Account Balance) for all days in month M for reseller R

Final Payout = Net Monthly Balance + Bonus

Note: If Net Monthly Balance is negative, the reseller owes the company.

A negative Final Payout shall require Admin review before confirmation.

Important: Separation of Company Money and Reseller Money

Company Due represents money owed to the bakery company for bread wholesale.
Amount Signed represents money the reseller takes back from their own cash return.
Account Balance is the net position after both the company is paid and the reseller takes their withdrawal.
These three quantities must never be conflated in system logic or user interface labelling.

Section 8 User Interface Requirements

8.1 General Principles

- The system shall be mobile-first: all screens shall be fully functional on a 360×640 dp viewport.
- Desktop usage shall be supported via a responsive layout with a minimum breakpoint of 1024 px width.
- The primary interaction model shall be tap-and-type on mobile; click-and-type on desktop.
- The colour language for financial states shall be consistent throughout: green for positive/settled, red for negative/owing, grey for neutral/pending.
- All currency values shall be formatted with the local currency symbol and thousands separators.

8.2 Depot Manager — Key Screens

Screen	Required Elements
Home / Dashboard	Today's date, count of resellers served vs. total active, quick-action buttons: 'New Distribution' and 'Record Payment', alert badge if anomalies exist
New Distribution	Reseller selector (searchable dropdown), date field (defaults to today), product line entries (product selector, quantity field, unit price field with default pre-filled), running total of expected company amount, Confirm Dispatch button
Record Payment	Reseller selector, distribution summary for the day (products, quantities, company due total), Amount Paid field, Amount Signed field (optional), real-time preview of Cash After Signing and Account Balance as values are entered, Submit button
Daily Summary	List of all resellers for the day: name, company due, amount paid, amount signed, account balance, colour-coded balance status
Alerts	List of active anomaly alerts with reseller name, alert type, date, and acknowledge button

8.3 Owner/Admin — Key Screens

Screen	Required Elements
Admin Dashboard	Overall daily totals, outstanding debts summary, anomaly count, monthly progress for all resellers
Reseller Management	Searchable/filterable list of resellers, add/edit/deactivate controls, view full history per reseller
Product Management	Product list with current prices, add/edit/discontinue controls, price history per product
Monthly Payouts	Month selector, table of all resellers with net monthly balance, bonus input field per reseller, calculated final payout, confirm payout button per reseller
Financial Reports	Date-range selector, report type selector (see Section 9), export to PDF/CSV button
Audit Log	Filterable table of all audit events, date range, user, action type filters, export button
Anomaly Review	Full list of alerts with context, acknowledge/dismiss controls

8.4 Mobile Input Constraints

- Quantity fields: numeric keyboard, integer only, minimum value 1
- Currency fields: numeric keyboard with decimal support, maximum 2 decimal places, minimum value 0
- Date fields: native date picker, default to current date
- Reseller and product selectors: searchable dropdown supporting type-ahead filtering

- All primary action buttons (Submit, Confirm, Save) shall be at least 48dp tall and positioned in the bottom half of the screen for thumb accessibility

Section 9 Reporting Requirements

All reports shall be accessible to the Admin role only and shall be exportable to PDF and CSV formats. Date ranges shall be selectable for all reports.

Report Name	Content	Filters Available
Daily Operations Report	All distributions and settlements for a given day: reseller, products, quantities, company due, amount paid, amount signed, account balance	Date, reseller, zone
Reseller Balance Statement	Daily breakdown of a single reseller's account balance across a date range, with running cumulative balance	Reseller, date range
Outstanding Debt Report	All resellers with a negative cumulative balance as of the selected date, ranked by debt amount	Date, zone, minimum debt threshold
Monthly Summary Report	Per-reseller: total distributions, total company due, total paid, total signed, net monthly balance, bonus, final payout	Month, year, reseller
Product Distribution Report	Quantity distributed per product per day across a date range; total expected revenue per product	Date range, product, reseller
Anomaly Report	All detected anomalies within a date range: type, reseller, date, description, resolution status	Date range, anomaly type, status
Audit Activity Report	All system actions within a date range filtered by user or action type	Date range, user, action type

Section 10 Alerts and Anomaly Detection Requirements

The system shall automatically detect and surface the following anomalies. All alerts shall be visible to both the Depot Manager and the Admin. Acknowledgement does not resolve the underlying issue — it records that the alert was reviewed.

#	Alert Name	Trigger Condition	Severity	Default Action
1	No Payment Recorded	Distribution exists for a reseller on date D but no settlement record has been submitted by end of day	High	Notify Admin; flag reseller on daily summary
2	Amount Signed Exceeds Amount Paid	Amount Signed > Amount Paid on a given settlement record	Critical	Block submission; display validation error to Depot Manager
3	Negative Daily Balance Exceeds Threshold	Account Balance < -(configurable threshold) for a single day	High	Alert Admin; log in anomaly report
4	Negative Cumulative Monthly Balance Exceeds Threshold	Net Monthly Balance < -(configurable threshold)	High	Alert Admin; flag reseller on monthly payout screen
5	Missing Distribution for Active Reseller	An active reseller has no distribution record for the current operational day by a configurable cut-off time	Medium	Notify Depot Manager
6	Price Override Without Justification	Depot Manager changes unit price at distribution entry without entering a reason note	Medium	Require reason field before saving; log override
7	Zero Quantity Distribution	A distribution record is submitted with quantity = 0	Medium	Warn Depot Manager; require explicit confirmation to proceed
8	Duplicate Distribution Record	A second distribution record is attempted for the same reseller and date	High	Block creation; notify Depot Manager; require Admin override
9	Negative Final Payout at Month End	Final Payout < 0 after bonus applied	High	Block payout confirmation; require Admin manual review and written justification

Configurable thresholds referenced above (items 3 and 4) shall be managed by the Admin via the system settings panel.

Section 11 Assumptions and Constraints

11.1 Business Assumptions

- A1: The company operates on a daily distribution cycle, 6 or 7 days per week. The system shall not enforce a maximum number of operating days.
- A2: Each reseller operates independently. One reseller's financial record does not affect another's.
- A3: The company does not accept return of unsold bread under any circumstance. This rule is immutable in v1.0.
- A4: The company price (wholesale price) is set by the Admin and represents the price at which bread is charged to resellers, not the retail price.
- A5: All monetary transactions are conducted in a single local currency. Multi-currency support is out of scope for v1.0.
- A6: A reseller may sell bread at any retail price of their choosing. The system does not capture retail prices.
- A7: Bonuses are discretionary, non-recurring, and set per reseller per month by the Admin. There is no automated bonus calculation rule.
- A8: The monthly financial period aligns with the calendar month (1st to last day).

11.2 Technical Constraints

- C1: The system shall be accessible via standard web browsers (Chrome 90+, Safari 14+, Firefox 88+) on mobile and desktop. A native app wrapper may be implemented in a future version.
- C2: Internet connectivity is required for full functionality. Limited offline support (data entry only) is required as specified in NFR-REL-03.
- C3: The system shall not integrate with external accounting or ERP systems in v1.0. Such integration shall be a future-version consideration.
- C4: All financial data shall be stored on a server controlled by the business owner. Cloud hosting is acceptable provided data residency requirements are met.
- C5: The system shall not send automated SMS or email notifications in v1.0. In-app alerts only.

11.3 Out of Scope for Version 1.0

- Inventory procurement and supplier management
- Payroll or employee salary management
- Customer-facing interfaces or retail point-of-sale
- Multi-branch or multi-depot management
- Automated SMS/email notification delivery
- Third-party accounting software integration
- Native mobile application (iOS / Android APK)

Document Control

Version 1.0 — This document is the baseline requirements specification. All subsequent changes must be tracked via a formal change request, reviewed by the Requirements Engineer, and approved by the Owner/Admin before implementation begins.